

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics
PRACTICAL NO - 12

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Practical 12: Analysis of Variance and Experimental Design

12. Analysis of Variance and Experimental Design:

a. Design and conduct an experiment to study the effects of different factors on a specific response variable, analyze the results using analysis of variance (ANOVA), and draw conclusions.

CODE:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

LOAD DATASET

```
file_path = "Retail and warehouse Sale - Retail and warehouse Sale.csv"
df = pd.read_csv(file_path)
print("Dataset loaded successfully!")
```

PART A – ANALYSIS OF VARIANCE (ANOVA)

1. Display first five records

```
print("\n1. First Five Records:")
print(df.head())
```

2. Number of rows and columns

```
print("\n2. Dataset Shape:")
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

3. Column names and data types

```
print("\n3. Column Names and Data Types:")
print(df.dtypes)
```

4. Missing values and duplicate records

```
print("\n4. Missing Values:")
print(df.isnull().sum())
print("\nNumber of Duplicate Records:")
print(df.duplicated().sum())
```

5. Different ITEM TYPE categories

```
print("\n5. Different ITEM TYPE Categories:")
print(df["ITEM TYPE"].unique())
print("\nNumber of ITEM TYPE Categories:")
print(df["ITEM TYPE"].nunique())
```

6. Average RETAIL SALES for each ITEM TYPE

```
average_sales = df.groupby("ITEM TYPE")["RETAIL SALES"].mean()
print("\n6. Average RETAIL SALES for Each ITEM TYPE:")
print(average_sales)
```

7. Box plot

```
plt.figure(figsize=(12, 6))

sns.boxplot(
    x="ITEM TYPE",
    y="RETAIL SALES",
    data=df
)
plt.title("Retail Sales by Item Type")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

8. Hypotheses

```
print("\n8. Hypotheses:")
print("H0: The average RETAIL SALES are equal for all ITEM TYPE categories.")
print("H1: At least one ITEM TYPE has a different average RETAIL SALES.")
```

9 & 10. One-way ANOVA

```
groups = [
    group["RETAIL SALES"].dropna()
    for name, group in df.groupby("ITEM TYPE")
]
f_statistic, p_value = f_oneway(*groups)
print("\n9 & 10. One-Way ANOVA Results:")
print("F-statistic:", f_statistic)
print("p-value:", p_value)
```

11 & 12. Compare p-value with 0.05

```
alpha = 0.05
print("\n11 & 12. Hypothesis Test:")

if p_value < alpha:
    print("p-value < 0.05")
    print("Reject the Null Hypothesis (H0).")
else:
    print("p-value >= 0.05")
    print("Do Not Reject the Null Hypothesis (H0).")
```

13. Conclusion

```
print("\n13. Conclusion:")
if p_value < alpha:
    print(
        "There is a statistically significant difference in average "
        "RETAIL SALES among the different ITEM TYPE categories."
    )
else:
```

```

print(
    "There is no statistically significant difference in average "
    "RETAIL SALES among the different ITEM TYPE categories."
)
print("Khushi T032")

```

OUTPUT:-

b. Implement a factorial experiment and analyze the main effects and interaction effects of factors using statistical techniques.

CODE:-

PART B – FACTORIAL EXPERIMENT

```

print("\n\n=====")
print("PART B – FACTORIAL EXPERIMENT")
print("=====")

```

14. Select ITEM TYPE and YEAR as factors

```

print("\n14. Factors:")
print("Factor 1: ITEM TYPE")
print("Factor 2: YEAR")
print("Response Variable: RETAIL SALES")

```

Check available years

```

print("\nAvailable Years:")
print(sorted(df["YEAR"].dropna().unique()))

```

```

print("\nNumber of Different Years:",
      df["YEAR"].nunique())

```

15. Average retail sales for each combination

```

combination_average = (
    df.groupby(["ITEM TYPE", "YEAR"])["RETAIL SALES"]
    .mean()
    .reset_index()
)
print("\n15. Average RETAIL SALES for Each ITEM TYPE and YEAR:")
print(combination_average)

```

Pivot table for better display

```

pivot_table = pd.pivot_table(
    df,
    values="RETAIL SALES",
    index="ITEM TYPE",
    columns="YEAR",
    aggfunc="mean"
)
print("\nAverage Sales Pivot Table:")
print(pivot_table)

```

16. Visualization

```

plt.figure(figsize=(12, 6))

sns.barplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES",
    hue="YEAR",
    estimator="mean",
    errorbar=None
)
plt.title("Average Retail Sales by Item Type and Year")
plt.xlabel("Item Type")
plt.ylabel("Average Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 17. Main effect of ITEM TYPE
print("\n17. Main Effect of ITEM TYPE:")

item_type_means = (
    df.groupby("ITEM TYPE")["RETAIL SALES"]
    .mean()
    .sort_values(ascending=False)
)
print(item_type_means)

# 18. Main effect of YEAR
print("\n18. Main Effect of YEAR:")
year_means = (
    df.groupby("YEAR")["RETAIL SALES"]
    .mean()
)
print(year_means)

# 19–22. Factorial ANOVA
print("\n19–22. Factorial ANOVA:")
if df["YEAR"].nunique() < 2:
    print("\nWARNING:")
    print("The dataset contains only one YEAR:",
          df["YEAR"].unique()[0])
    print(
        "\nFactorial ANOVA cannot be performed using YEAR "
        "because at least two different YEAR levels are required."
    )
    print("\nTherefore:")
    print("ITEM TYPE main effect: Can be analyzed.")
    print("YEAR main effect: Cannot be tested statistically.")
    print("ITEM TYPE × YEAR interaction: Cannot be tested.")
else:

    # Two-way factorial ANOVA
    model = ols(

```

```

'Q("RETAIL SALES") ~ C(Q("ITEM TYPE")) * C(YEAR)',
data=df
).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print("\n20. Factorial ANOVA Results:")
print(anova_table)
print("\n21. P-values:")
for effect in anova_table.index:
    print(effect, "p-value =",
          anova_table.loc[effect, "PR(>F)"])

# 22. Identify significant effects
print("\n22. Significance at alpha = 0.05:")
for effect in anova_table.index:
    if effect == "Residual":
        continue
    p = anova_table.loc[effect, "PR(>F)"]
    if p < 0.05:
        print(effect, "-> Significant")
    else:
        print(effect, "-> Not Significant")

# 23. Final Business Conclusion
print("\n23. FINAL BUSINESS CONCLUSION:")
if p_value < 0.05:
    print(
        "The one-way ANOVA indicates that ITEM TYPE has a "
        "significant effect on RETAIL SALES. Therefore, "
        "retail sales vary significantly among different "
        "item categories."
    )
else:
    print(
        "The one-way ANOVA indicates that ITEM TYPE does not "
        "have a statistically significant effect on RETAIL SALES."
    )

if df["YEAR"].nunique() < 2:
    print(
        "The uploaded dataset contains only the year 2020. "
        "Hence, the effect of YEAR and the ITEM TYPE × YEAR "
        "interaction cannot be statistically evaluated."
    )
else:
    print(
        "The factorial ANOVA evaluates the effects of ITEM TYPE, "
        "YEAR, and their interaction on RETAIL SALES."
    )
print("Khushi T032")

```

OUTPUT:-